

π_0 · arXiv 2410.24164 · RSS 2025

Visionary

11+3 · seat 3

2026 BEHAVIOR defaults
VLM semantic prior + generative chunks

Sep 28 paper club · VLA Architectures

Spine: PaliGemma + OT-CFM action chunks (DDPM→flow)

Physical Intelligence · H=50 · ~10 Euler · ~73 ms RTX4090 · NOT a classical planner

Visionary arc



2025 winners did not replace the substrate — they specialized noise + overlap.

2026: FM+chunk is the default control prior; hierarchy/stage memory is the bottleneck.

2026 BEHAVIOR defaults

FM + chunk prior

2025 winners specialized
 Σ / overlap — not the head

Start pi05_base

Open weights > closed demos
for challenge forks

Deploy with RTC

Train FM chunks;
async soft-inpaint at run

Hierarchy next

Stage memory / System-2
> another -0.5 FID of FM

Port '25 ideas

corr Σ · $N=15$ · soft inpaint
exec26/keep4 · stages

Durable idea

VLM semantic prior

+

generative continuous chunks

Head may shift: FM · diffusion · FAST+flow hybrid

Chunked continuous control interface will not.

No Discord / Drive digressions — stay on 2026 control bets

What to try for BEHAVIOR'26

1

Fork openpi pi05_base

2

Port corr Σ + multi-sample FM

3

Soft-inpaint horizons / RTC

4

Stage embeddings / memory

5

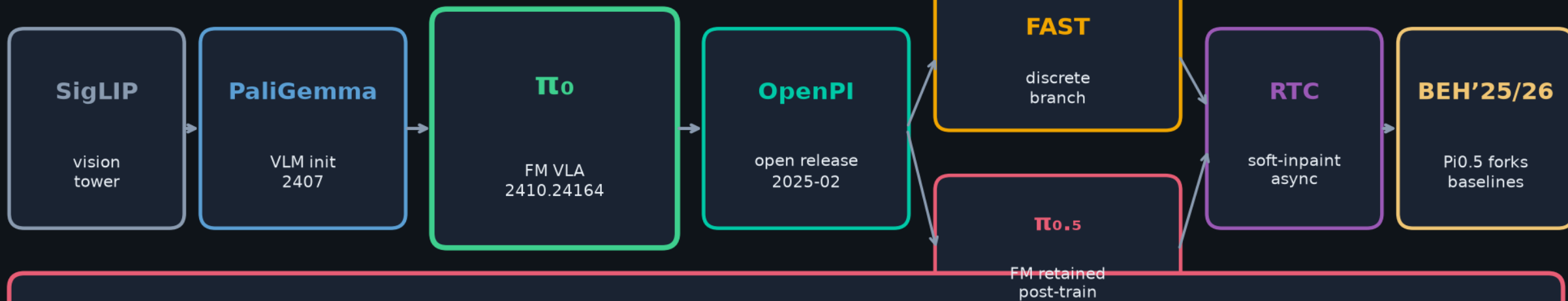
Dual-track vs GR00T N1.7

6

Ablate on 7-scene split

Genealogy · CLEAR club line

SigLIP → PaliGemma → π_0 → OpenPI → (FAST $\parallel$ $\pi_{0.5}$) → RTC → BEHAVIOR'25 → '26



CRITICAL CORRECTION

$\pi_{0.5}$ retains flow matching at post-train (OpenPI + $\pi_{0.5}$ paper)

Secondary blogs claiming “FAST replaced FM” = FALSE · hybrid FAST pretrain + FM post-train

Pivot this paper locks

FM VLA + VLM init + action expert + 50 Hz chunks at ~10k h cross-embodiment scale

Orthogonal control knobs

First-class for 2026 · not just the VLM backbone

H

train horizon

paper default 50
sweep {8,16,25,50}

T_a / K

open-loop commit

16@20Hz · 25@50Hz
mid ≈ 0.5 –1 s sweet

NFE

Euler steps

paper ~ 10
saturates ~ 5 –10

async

overlap / RTC

soft-inpaint
exec26 / keep4

Σ

noise corr.

isotropic $\rightarrow$
 $0.5\Sigma + 0.5I$

Latency budget (Table I · RTX 4090 · 3 cams)

Image enc

14 ms

Obs forward

32 ms

$\times 10$ FM

27 ms

On-board

73 ms

+Wi-Fi

~ 86 ms

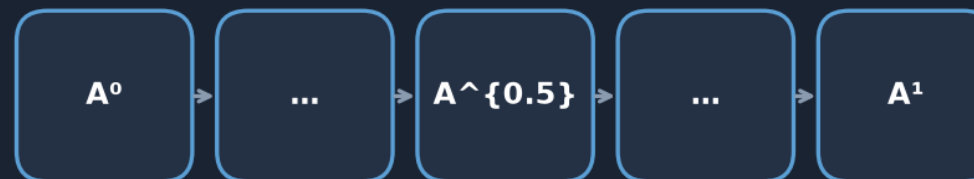
OT-CFM action chunks · DDPM → flow

Linear path $A^\tau = \tau A + (1-\tau)\varepsilon$ · target $u = A - \varepsilon$ · $\tau \sim \text{shifted Beta}(1.5, 1)$

Straight OT path in action space



Inference: ~10 Euler steps



$$A^{\{\tau+\delta\}} = A^\tau + \delta \cdot v_\theta(A^\tau, o) \quad \cdot \quad \delta=0.1$$

Chunk → commit → replan (DP/ACT pattern; FM head)

Train H=50

propose full chunk

Execute K

16@20Hz / 25@50Hz

Open-loop

no temporal ensemble

Replan

new infer; RTC later

Architecture · PaliGemma + OT-CFM action expert

~3.3B total · MoE dual-stream (Transfusion-inspired) · blockwise causal attention

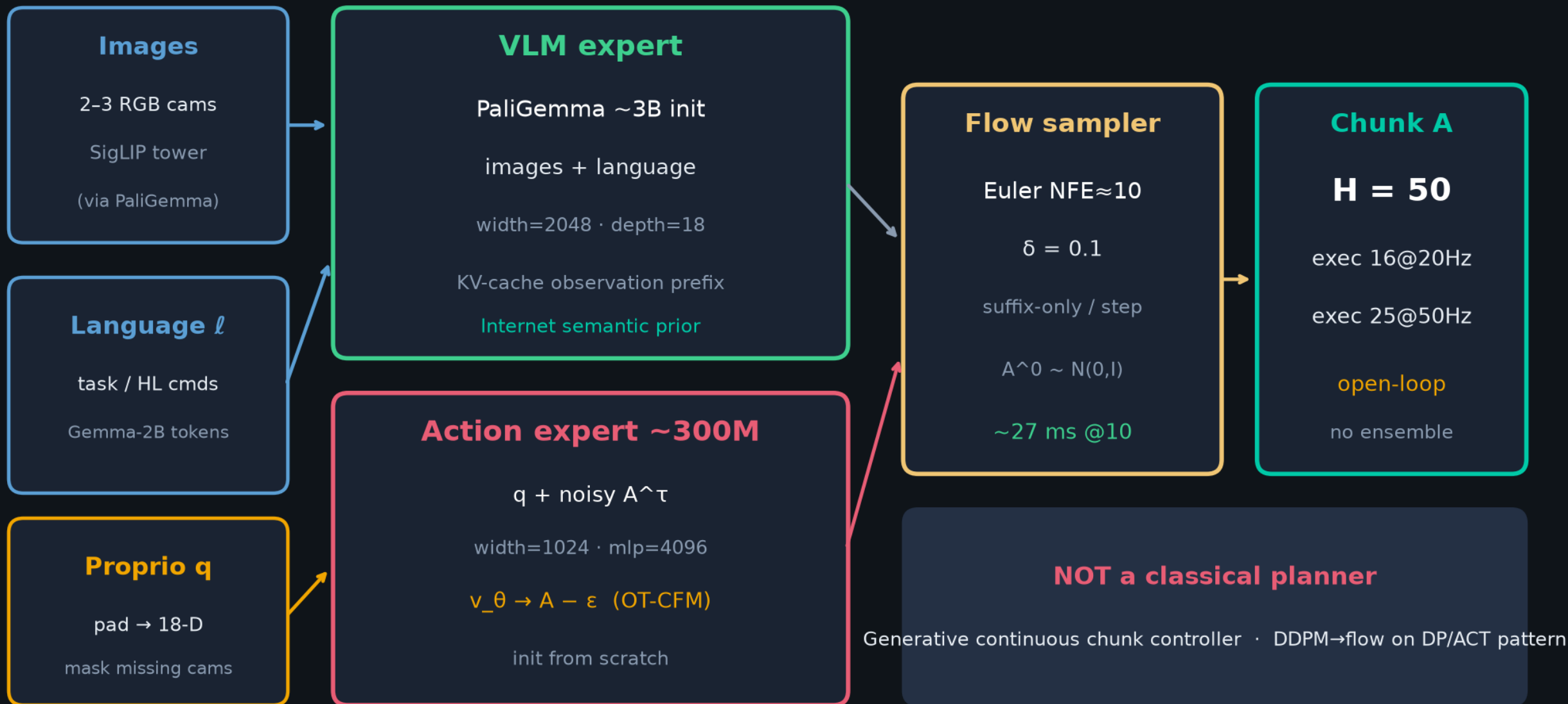




Fig. 12: We evaluate a range of complex and temporally extended tasks. This includes: folding laundry from a bin with a stationary (a) or mobile (b) robot, bussing a real lunch table (c), assembling a box (d), packing eggs into a carton (e), and packing food into a to-go box (f). These tasks require combining dozens of individual behaviors, such as grasping, stacking, folding, and flattening, generalization to a huge variety of object configurations, and complex physical properties, such as deformable objects or flexible cardboard.

are presented in Figure 13. The scoring rubrics are in Appendix E. A score of 1.0 represents a perfect execution, while partial scores correspond to partially completed tasks (e.g., 0.5 indicates that half the objects were bussed correctly). These tasks are very difficult, and we were not able to solve them with other methods. We therefore use these tasks to compare to ablations of our approach, evaluating π_0 after pre-training and fine-tuning, out of the box after pre-training only (“out-of-box”), and training on the fine-tuning data without any pre-training (“scratch”). The results show that π_0 can solve many of these tasks, with our full pre-training and fine-tuning recipe performing best across the board. Note that many of these more difficult tasks show a very large improvement from using the pre-trained model, indicating that pre-training is especially useful with harder tasks. The absolute performance of π_0 varies across the tasks, likely due to differences in task difficulty and the degree to which the tasks are represented in pre-training. We recommend that readers watch the task videos on the [accompanying website](#) for a more complete impression of these tasks and their complexity. We believe that this level of autonomous performance on such challenging

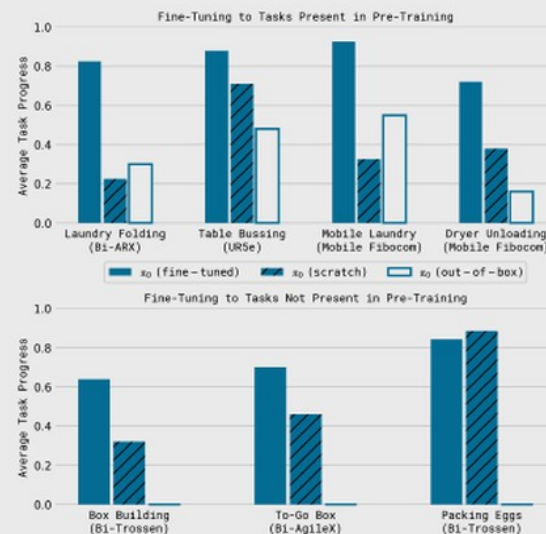


Fig. 13: Post-training results on complex tasks in terms of average scores over 10 trials. The full pre-trained π_0 model attains more than 50% of the maximum score across all of the tasks, and typically outperforms the ablations, with especially significant improvements on the hardest tasks.

Our empirical evaluation studies tasks that combine dexterity, generalization, and temporally extended multi-stage behaviors. Our model incorporates Internet-scale vision-language model (VLM) pre-training with flow matching for representing complex high-frequency action chunks. Our pre-training mixture consists of 10,000 hours of dexterous manipulation data from 7 different robot configurations and 68 tasks, in addition to large amounts of previously collected robot manipulation data from OXE [10], DROID [23], and Bridge [52]. To our knowledge, this represents the largest pre-training mixture ever used for a robot manipulation model. Our fine-tuning experiments include over 20 tasks, where we show that our model outperforms a variety of baselines, including prior VLA models [24] and models designed specifically for dexterous manipulation [57, 9]. We also examine how our post-training recipe can enable highly complex tasks, such as folding multiple articles of clothing from arbitrary initial configurations or assembling boxes.

Our framework broadly resembles the training procedures employed for large language models, which typically consist

Visionary · close

Durable idea: VLM semantic prior + generative continuous chunks.

Start pi05_base · deploy with RTC · skip Discord/Drive

Q&A · Sep 28 · π_0 / 2410.24164

π_0 : A Vision-Language-Action Flow Model for General Robot Control

Physical Intelligence

Kevin Black, Noah Brown, Danny Driess, Adnan Esmail, Michael Equi, Chelsea Finn, Niccolo Fusai, Lachy Groom, Karol Hausman, Brian Ichter, Szymon Jakubczak, Tim Jones, Liyiming Ke, Sergey Levine, Adrian Li Bell, Mohith Mohukuri, Suraj Nair, Karl Pertsch, Lucy Xiaoqiang Shi, James Tapper, Quan Vuong